

Quantitative Aptitude

Number System

STUDY NOTES

Remainder Theorem



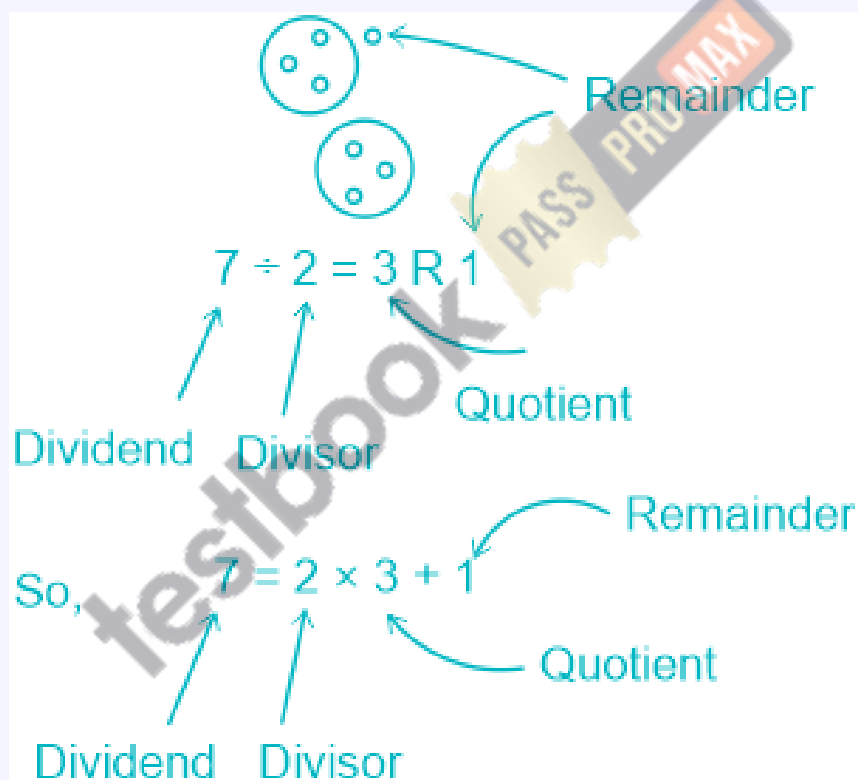
Remainder Theorem and Factor Theorem

Remainder:

The remainder is the value left after the division if the divisor does not completely divide the dividend.

If the divisor completely divides the dividend, then the remainder will be zero.

Method of Division in real life:



Example 1: A number when divided by 1147 the remainder is 61. What will be the remainder obtained by dividing the same number by 37.

Solution:

Given:

Divisor = 1147

Remainder = 61

Concept:

If the multiple of any number divides another number then the number also divides another number

Dividend = Divisor $\times$ quotient + Remainder

Calculations:

Dividend = Divisor $\times$ quotient + Remainder

$\Rightarrow$ Dividend = $1147 \times q + 61$

$\Rightarrow (37 \times 31)q + 37 + 24$

$\Rightarrow 37(31q + 1) + 24$

When this number is divided by then 24 will be the remainder

$\Rightarrow$ Remainder = 24

$\therefore$ The correct answer is 24.

Long Division Method:

Similarly when we divide a polynomial:

$f(x) \div d(x) = q(x)$ with a remainder of $r(x)$

which we can write as:

$$f(x) = d(x) \cdot q(x) + r(x)$$

$\uparrow$
Dividend
 $\uparrow$
Divisor
 $\uparrow$
Quotient
 $\uparrow$
Remainder

Example 2: If divide the polynomial $x + 3x^2 - 1$ by $d(x) = x + 1$, the remainder is:

Solution:

- $f(x)$ is $x + 3x^2 - 1$
- $d(x)$ is $x + 1$

Step 1: We can express the dividend and divisor in standard form by arranging their terms in descending order of their degrees. Thus, the dividend becomes $3x^2 + x - 1$ and the divisor becomes $x + 1$.

Step 2: We divide the first term of the dividend $3x^2$ by the first term of the divisor, i.e., we divide $3x^2$ by x and get $3x$. This gives us the first term of the quotient.

$$3x^2 / x = 3x = \text{first term of quotient}$$

Step 3: We begin the division process by multiplying the divisor $(x + 1)$ by the first term of the quotient $(3x)$. The resulting product $(3x^2 + 3x)$ is then subtracted from the dividend $(3x^2 + x - 1)$. This leaves us with a remainder of $-2x - 1$.

$$\begin{array}{r} 3x \\ x+1 \overline{) 3x^2 + x - 1} \\ \underline{3x^2 + 3x} \\ -2x - 1 \end{array}$$

Step 4: We now consider the remainder $(-2x - 1)$ as the new dividend, while the divisor remains unchanged. Following Step 2, we divide the first term of the new dividend $(-2x)$ by the first term of the divisor (x) , resulting in -2 . This value (-2) becomes the second term of the quotient.

$$-2x / x = -2 = \text{second term of quotient, So new Quotient} = 3x - 2$$

Step 5: We proceed by multiplying the divisor $(x + 1)$ by the second term of the quotient (-2) . The resulting product $(-2x - 2)$ is then subtracted from the dividend $(-2x - 1)$, leaving us with a remainder of 1 .

$$\begin{array}{r|l}
 (x+1)(-2) & -2x-1 \\
 = -2x-2 & -2x-2 \\
 & + \quad + \\
 & \hline
 & +1
 \end{array}$$

We continue this division process until either the remainder becomes 0 or the degree of the new dividend is less than the degree of the divisor. When this condition is met, the remaining dividend becomes the final remainder, and the sum of all the quotients we have obtained gives us the complete quotient.

Step 6: Thus, the quotient in full is $3x - 2$ and the remainder is 1.

Let us look at what we have done in the process above as a whole:

$$\begin{array}{r}
 3x-2 \\
 \hline
 x+1 \overline{) 3x^2 + x - 1} \\
 \underline{3x^2 + 3x} \\
 -2x - 1 \\
 \underline{-2x - 2} \\
 + \quad + \\
 \hline
 1
 \end{array}$$

Notice that $3x^2 + x - 1 = (x + 1)(3x - 2) + 1$

i.e, Dividend = (Divisor × Quotient) + Remainder

Remainder = 1

∴ **The correct answer is 1.**

Remainder Theorem:

$$\begin{array}{c}
 \text{Divident} \\
 \uparrow \\
 \text{Divisor} \rightarrow x) \overline{y} \quad (z \rightarrow \text{Quotient} \\
 \quad \quad \quad \underline{a} \rightarrow \text{Remainder}
 \end{array}$$

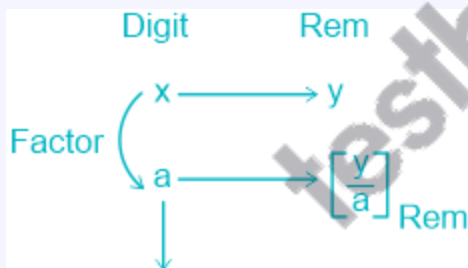
$$\text{Dividend} = \text{Divisor} \times \text{Quotient} + \text{Remainder}$$

When dividing a number by divisor 'N', the remainder is always smaller than N.

When we simplify a fraction by any digit to find the remainder, then in the end we have to multiply the simplified digit by the actual digit to get the actual remainder.

Example:

$$\begin{array}{l}
 \frac{24}{7} = \text{Rem [3]} \\
 \swarrow \quad \searrow \\
 8 \quad \quad \times 8 \\
 \downarrow \quad \quad \downarrow \\
 \frac{192}{7} = \frac{24}{7} = \text{Rem [3]}
 \end{array}$$



If a is not a factor then the answer can not be determined.

Example 3: On dividing the polynomial. $f(x) = x^3 - 5x^2 + 6x - 3$ by a polynomial $g(x)$, the quotient $q(x)$ and remainder $r(x)$ are $(x - 1)$, (-1) , respectively, find the polynomial $g(x)$.

Solution:

As we know dividend = (divisor $\times$ quotient) + remainder

$$\text{Dividend} = x^3 - 5x^2 + 6x - 3$$

$$\text{Divisor} = g(x)$$

$$\text{Quotient} = x - 1$$

$$\text{Remainder} = -1$$

Putting in the formula,

$$\Rightarrow x^3 - 5x^2 + 6x - 3 = (g(x) \times (x - 1)) + (-1)$$

$$\Rightarrow x^3 - 5x^2 + 6x - 3 + 1 = (g(x) \times (x - 1))$$

$$\Rightarrow x^3 - 5x^2 + 6x - 2 = (g(x) \times (x - 1))$$

$$\Rightarrow (x^3 - 5x^2 + 6x - 2)/(x - 1) = g(x)$$

$$\Rightarrow g(x) = x^2 - 4x + 2$$

$\therefore$ The correct answer is $g(x) = x^2 - 4x + 2$.

Theorem:

When we divide a polynomial $f(x)$ by $(x-c)$ the remainder is $f(c)$.

When we divide $f(x)$ by $(x - c)$, the equation looks like this:

$$f(x) = (x - c) q(x) + r$$

Since $x - c$ is a simple polynomial with degree 1, the remainder r is just a constant number.

Now, if we replace x with c in the equation, it becomes:

$$f(c) = (c - c)q(c) + r$$

This simplifies to:

$$f(c) = 0 \times q(c) + r$$

$$f(c) = r$$

So, the value of $f(c)$ is simply the remainder, r .

Example 4: Find the remainder when $2x^2 - 5x - 1$ is divided by $x - 5$.

Solution:

Here, $f(x) = 2x^2 - 5x - 1$

$$x - 5 = 0 \Rightarrow x = 5$$

To find the remainder, calculate $f(5)$

$$\text{So, } f(5) = 2(5)^2 - 5(5) - 1$$

$$f(5) = 2(25) - 25 - 1 = 50 - 25 - 1 = 24$$

The remainder obtained if we divide $f(x) = 2x^2 - 5x - 1$ by $x - 5$ is 24.

$\therefore$ The correct answer is 24.

Example 5: Find the remainder, when we divide $2x^3 - 3x^2 + 6x - 4$ by $2x - 3$.

Solution:

Given:

$$2x^3 - 3x^2 + 6x - 4$$

Divisor: $2x - 3$

Formula used:

Remainder Theorem: If a polynomial $f(x)$ is divided by $(ax - b)$, the remainder is $f(b/a)$

Calculation:

$$\text{Let } f(x) = 2x^3 - 3x^2 + 6x - 4$$

Divisor: $2x - 3$

According to the Remainder Theorem, $f(3/2)$ will give the remainder.

$$\Rightarrow f(3/2) = 2(3/2)^3 - 3(3/2)^2 + 6(3/2) - 4$$

$$\Rightarrow f(3/2) = 2(27/8) - 3(9/4) + 9 - 4$$

$$\Rightarrow f(3/2) = 27/4 - 27/4 + 9 - 4$$

$$\Rightarrow f(3/2) = 0 + 5$$

$$\Rightarrow f(3/2) = 5$$

$\therefore$ The correct answer is 5.

Example 6: Let m be a non-zero integer and n be a positive integer. Let R be the remainder obtained on dividing the polynomial $x^n + m^n$ by $(x - m)$. Then R is:

Solution:

Concept :

Remainder Theorem:

The remainder theorem states that when a polynomial $p(x)$ is divided by a linear polynomial $(x - a)$, then The remainder is equal to $p(a)$.

Calculation:

$$\text{Let } P(x) = x^n + m^n$$

If $P(x)$ is divided by $(x - m)$.

By using the remainder theorem, the remainder

$$\Rightarrow R = P(m) = m^n + m^n$$

$$\Rightarrow R = 2m^n$$

$\therefore$ The correct answer is R is a non-zero even integer.

Factor Theorem:

- When $f(c) = 0$ then $x - c$ is a factor of $f(x)$
- When $x - c$ is a factor of $f(x)$ then $f(c) = 0$

Successive Division:

When a number N is divided by a , the remainder is c . If we then divide this quotient by b , the remainder is d . This process is known as successive division.

$$\begin{array}{r} a) \overline{V} \begin{array}{l} x \\ X \\ \hline C \end{array} \\ a) \overline{x} \begin{array}{l} x \\ X \\ \hline d \end{array} \end{array}$$

Example 7: After dividing a number successively by 2, 3 and 5, the remainder are 1, 2 and 3, respectively. What will be the remainder, if 13 divides the same number (if the last quotient is 1)?

Solution:

Dividend = Divisor $\times$ Quotient + Remainder

The number X in this case will be determined as follows:

2	X	1
3	Y	2
5	Z	3
	1	

We know that, Dividend = Divisor $\times$ Quotient + Remainder

$$Z = (5 \times 1) + 3 = 8$$

$$Y = (3 \times Z) + 2 = (3 \times 8) + 2 = 26$$

$$X = (2 \times Y) + 1 = (2 \times 26) + 1 = 53$$

$$\therefore X = 53$$

According to the question,

When (X) is divided by 13 i.e. $53/13$ then the remainder will be 1.

$\therefore$ The correct answer is 1.

Concept of Negative Remainder:

$$75 \div 17 = \text{Rem } [7 \text{ or } -10]$$

When we divide a number by 100, then we get the last two digits as a remainder.

Ex: $63192 \div 100 = \text{Rem } [92]$

When we divide a number by 1000, then we get the last three digits as a remainder.

Ex: $63192 \div 1000 = \text{Rem } [192]$

Example 8: What will be the remainder when $31^{13} + 41^{13}$ is divided by 5

Solution:

Given:

$$31^{13} + 41^{13}$$

Theorem used:

Factors theorem

$(a^n + b^n)$ is divisible by $(a + b)$, if n is odd

Calculation:

Using the factor theorem

$$31^{13} + 41^{13} \text{ is divisible by } (31 + 41) = 72$$

When 72 is divided by 5 it gives the remainder 2.

$\therefore$ The correct answer is 2.

Example 9: If we divide $x^4 - y^4$ by $x - y$, then the quotient and remainder are respectively

Solution:

Concept: Dividend = Divisor Quotient + Remainder

Calculation:

Given: Divisor = $x - y$,

Dividend = $x^4 - y^4$

$$\begin{array}{r}
 x-y \overline{) x^4 - y^4} \quad (x^3 + x^2y + xy^2 + y^3) \\
 \underline{x^4 \quad - x^3y} \\
 0 \quad -x^3y - y^4 \\
 \underline{+x^3y \quad -x^2y^2} \\
 0 \quad -x^2y^2 - y^4 \\
 \underline{+x^2y^2 \quad -xy^3} \\
 0 \quad -xy^3 - y^4 \\
 \underline{+xy^3 + y^4} \\
 0
 \end{array}$$

∴ The correct answer is Quotient = $x^3 + x^2y + xy^2 + y^3$, and remainder = 0
